



Beckmann, M., Beinert, R., and Bresch, J. (2025) – Max-normalized Radon CDT for limited data classification, *Lect. Notes Comput. Sci.*

Beckmann, M., Beinert, R., and Bresch, J. (2026) – Normalized Radon CDT for invariance and robustness in optimal transport based image classification, *SIAM J. Imaging Sci.*

Beckmann, M., Beinert, R., and Bresch, J. (2026) – Generalization of the normalized Radon CDT for limited data recognition, *J. Math. Imaging Vis.*

Bresch, J. (2026) – Geometry-aware Ansätze in Image Processing and Stochastic Calculations of Matrix Quantities, *Dissertation*

Generalization of Normalized Radon Cumulative Distribution Transforms

in the Light as Feature Extractors for Image Classification

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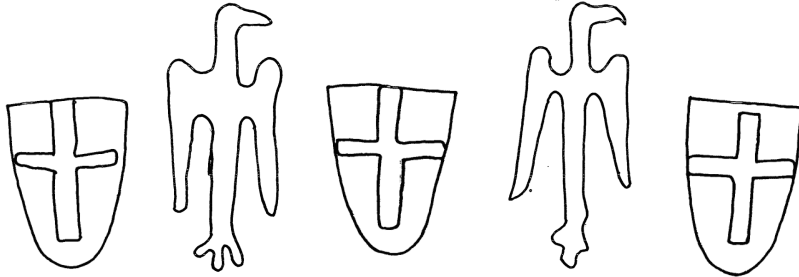
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- Automatic classification of images, for instance watermarks in historical manuscripts^[1] (see below).
- **Challenges:** hundreds of classes, few samples per class.
- Reasonable assumption: “*affine*” class structure.
- Need **invariance** to translations/dilations and, generally, stability under (non-)affine deformations^{[2][3]}.



[1] Hauser, D. et al. (2024) – “On image processing and pattern recognition for thermograms of watermarks in manuscripts – A first proof-of-concept,

[2] Wang, W. et al. (2013) – A linear optimal transportation framework for quantifying and visualizing variations in sets of images, *Int. J. Comput. Vis.*

[3] Moosmüller, C. and Cloninger, A. (2023) – Linear optimal transport embedding: Provable Wasserstein classification for certain rigid transformations and perturbations, *IMA J. Appl. Mat.*

- Feature extractor via Radon cumulative distribution transform (R-CDT) ^[4]
 - based on one-dimensional optimal transport maps (CDTs), and generalized to
 - $\mathbb{R}^2$, via the 2-dimensional Radon transform for R-CDT for $\mathcal{P}(\mathbb{R}^2)$,
 - $\mathbb{R}^d$, via the d -dimensional Radon transform for R-CDT for $\mathcal{P}(\mathbb{R}^d)$,
 - “more” general data on $\mathbb{X}$, via the *generalized, circular and rotation Radon transform* ^[5] for R-CDT for $\mathcal{P}(\mathbb{X})$.
- Introduce h -normalized Radon-CDT (h NR-CDT) via two-step normalization.
- *Linear separability* for
 - affinely transformed measure classes → single-point sets in h NR-CDT-space.
 - distorted measure classes → balls in h NR-CDT-space.
- Classifier: nearest template and neighbour, (linear) SVM.
- Numeric: on synthetic, LinMNIST ^[6], Chinese MNIST ^[7], “ABC” ^[8], “FAUST” ^[9], and rotation SO(3) dataset.
- Comparison: with LOT ^{[2][3]}, R-CDT ^[4], momentum-based (MB) methods ^[10], and neuronal networks (NN).

[4] Kolouri, S., Park, S. R., and Rohde, G. K. (2016) – The Radon cumulative distribution transform and its application to image classification, *IEEE Trans. Image Process.*

[5] Natterer, F. (2001) – The Mathematics of Computerized Tomography,

[6] Deng, L. (2012) – The MNIST database of handwritten digit images for machine learning research, *IEEE Signal Process. Mag.*

[7] Liu, C.-L. et al. (2011) – “CASIA online and offline Chinese handwriting databases,

[8] Koch, S. et al. (2019) – “ABC: A big CAD model dataset for geometric deep learning,

[9] Bogo, F. et al. (2014) – “FAUST: Dataset and evaluation for 3D mesh registration,

[10] Flusser, J. and Suk, T. (1993) – Pattern recognition by affine moment invariants, *Pattern Recognit.*

- For $\theta \in \mathbb{S}_1 := \{\mathbf{x} \in \mathbb{R}^2 \mid \|\mathbf{x}\| = 1\}$, define the slicing operator

$$S_\theta: \mathbb{R}^2 \rightarrow \mathbb{R}, \quad \mathbf{x} \mapsto \langle \mathbf{x}, \theta \rangle, \quad \mathbf{x} \in \mathbb{R}^2.$$

- with inverse $S_\theta^{-1}(t) = \{t\theta + \tau\theta^\perp \mid \tau \in \mathbb{R}\} \subset \mathbb{R}^2$.
- Let $f \in L^1(\mathbb{R}^2)$, its *Radon transform* $\mathcal{R}[f]: \mathbb{R} \times \mathbb{S}_1 \rightarrow \mathbb{R}$ is the line integral, where ds is the arc length of $S_\theta^{-1}(t)$

$$\mathcal{R}[f](t, \theta) := \int_{S_\theta^{-1}(t)} f(s) ds, \quad (t, \theta) \in \mathbb{R} \times \mathbb{S}_1,$$

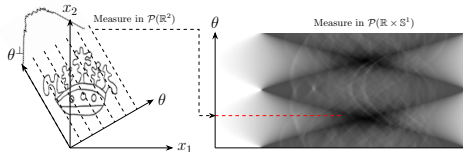
- For direction $\theta \in \mathbb{S}_1$, the *restricted Radon transform* $\mathcal{R}_\theta$ of signed, regular, finite measures $\mu \in \mathcal{M}(\mathbb{R}^2)$ is

$$\mathcal{R}_\theta: \mathcal{M}(\mathbb{R}^2) \rightarrow \mathcal{M}(\mathbb{R}), \quad \mu \mapsto (S_\theta)_\# \mu = \mu \circ S_\theta^{-1},$$

- the *Radon transform* $\mathcal{R}: \mathcal{M}(\mathbb{R}^2) \rightarrow \mathcal{M}(\mathbb{R} \times \mathbb{S}_1)$ via $\mathcal{R}[\mu] := \mathcal{I}_\#[\mu \times u_{\mathbb{S}_1}]$ with

$$\mathcal{I}(\mathbf{x}, \theta) := (S_\theta(\mathbf{x}), \theta) \quad (\mathbf{x}, \theta) \in \mathbb{R}^2 \times \mathbb{S}_1.$$

- Notice: $\mathcal{R}[f\lambda_{\mathbb{R}^2}] = \mathcal{R}[f]\sigma_{\mathbb{R} \times \mathbb{S}_1}$ for all $f \in L^1(\mathbb{R}^2)$ w.r.t. the surface measure on $\mathbb{R} \times \mathbb{S}_1$.



Proposition (Disintegration, Connection)

Let $\mu \in \mathcal{M}(\mathbb{R}^2)$. For $g \in C_0(\mathbb{R} \times \mathbb{S}_1)$, $h \in L^\infty(\mathbb{R} \times \mathbb{S}_1)$ holds

$$\langle \mathcal{R}[\mu], g \rangle = \int_{\mathbb{S}_1} \langle \mathcal{R}_\theta[\mu], g(\cdot, \theta) \rangle d u_{\mathbb{S}_1}(\theta)$$

$$\langle \mathcal{R}[\mu], h \rangle = \langle \mu, \mathcal{R}^*[h] \rangle, \quad \mathcal{R}^*[h] := \int_{\mathbb{S}_1} h(S(\cdot, \theta), \theta) d u_{\mathbb{S}_1}(\theta)$$

Proposition (Affinely Transformed Measures)

For any $\theta \in \mathbb{S}_1$, $\mathbf{A} \in \text{GL}(2)$ and $\mathbf{y} \in \mathbb{R}^2$, and $\mu_{\mathbf{A}, \mathbf{y}} := (\mathbf{A} \cdot + \mathbf{y})_\# \mu = \mu \circ (\mathbf{A}^{-1}(\cdot - \mathbf{y}))$ holds

$$\mathcal{R}_\theta[\mu_{\mathbf{A}, \mathbf{y}}] = (\|\mathbf{A}^\top \theta\| \cdot + \langle \mathbf{y}, \theta \rangle)_\# \mathcal{R}_{\mathbf{A}^\top \theta / \|\mathbf{A}^\top \theta\|}[\mu]$$

$$= \mathcal{R}_{\mathbf{A}^\top \theta / \|\mathbf{A}^\top \theta\|}[\mu] \circ \left(\frac{\cdot - \langle \mathbf{y}, \theta \rangle}{\|\mathbf{A}^\top \theta\|} \right).$$

transformation	$\mathbf{A}$	$\mathbf{y}$	$\mathcal{R}_{\theta(\vartheta)}[\mu_{\mathbf{A}, \mathbf{y}}]$, $\vartheta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ with $\theta(\vartheta) := (\cos(\vartheta), \sin(\vartheta))^\top$
translation	$\mathbf{I}$	$\mathbb{R}^2$	$\mathcal{R}_{\theta(\vartheta)}[\mu] \circ (\cdot - \langle \mathbf{y}, \theta(\vartheta) \rangle)$
rotation	$\begin{pmatrix} \cos(\varphi) & -\sin(\varphi) \\ \sin(\varphi) & \cos(\varphi) \end{pmatrix}$	$\mathbf{0}$	$\mathcal{R}_{\theta(\vartheta - \varphi)}[\mu]$
reflection	$\begin{pmatrix} \cos(\varphi) & \sin(\varphi) \\ \sin(\varphi) & -\cos(\varphi) \end{pmatrix}$	$\mathbf{0}$	$\mathcal{R}_{\theta(\varphi - \vartheta)}[\mu]$
anisotropic scaling	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	$\mathbf{0}$	$\mathcal{R}_{\theta(\arctan(\frac{b}{a} \tan(\vartheta)))}[\mu] \circ ([a^2 \cos^2(\vartheta) + b^2 \sin^2(\vartheta)]^{-1/2} \cdot)$
vertical shear	$\begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}$	$\mathbf{0}$	$\mathcal{R}_{\theta(\arctan(c + \tan(\vartheta)))}[\mu] \circ ([1 + c^2 \cos^2(\vartheta) + c \sin(2\vartheta)]^{-1/2} \cdot)$

- Let $\mathbb{X}$ be a *Polish space*, i.e. a separable completely metrizable topological space, and let $\mathcal{M}(\mathbb{X})$ denotes the set of finite, signed, and regular (Borel) measures and $\mathcal{P}(\mathbb{X})$ denotes the set of probability measures.
- The *generalized slicing operator* with the *direction set* Θ and measurable *defining function* $\phi: \mathbb{X} \times \Theta \rightarrow \mathbb{R}$ is defined by

$$S_{\phi, \theta}: \mathbb{X} \rightarrow \mathbb{R}, \quad \mathbf{x} \mapsto \phi(\mathbf{x}, \theta), \quad \theta \in \Theta,$$

- The *generalized restricted Radon transform* as

$$\mathcal{R}_{\phi, \theta}: \mathcal{M}(\mathbb{X}) \rightarrow \mathcal{M}(\mathbb{R}), \quad \mu \mapsto (S_{\phi, \theta})_{\#} \mu := \mu \circ S_{\phi, \theta}^{-1}.$$

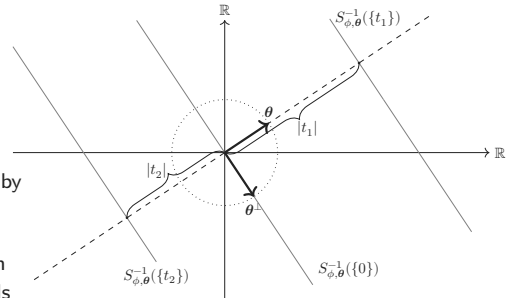
- The *gluing measure* $\gamma \in \mathcal{P}(\Theta)$ and *gluing map*

$$\mathcal{I}_{\phi}: \mathbb{X} \times \Theta \rightarrow \mathbb{R} \times \Theta, \quad (\mathbf{x}, \theta) \mapsto (S_{\phi, \theta}(\mathbf{x}), \theta).$$

- The *generalized Radon transform* if Θ is a Polish space, too, by

$$\mathcal{R}_{\phi, \gamma}: \mathcal{M}(\mathbb{X}) \rightarrow \mathcal{M}(\mathbb{R} \times \Theta), \quad \mu \mapsto (\mathcal{I}_{\phi})_{\#} [\mu \times \gamma].$$

- Restricting the domain of the generalized (restricted) Radon transform to probability measures on $\mathbb{X}$ yields a map towards probability measures on $\mathbb{R} \times \Theta$ (respectively $\mathbb{R}$).



Proposition (Disintegration, Connection)

Let $\mu \in \mathcal{M}(\mathbb{X})$. Then for all continuous functions $g \in C_0(\mathbb{R} \times \Theta)$ vanishing at infinity holds

$$\langle \mathcal{R}_{\phi, \gamma}[\mu], g \rangle = \int_{\Theta} \langle \mathcal{R}_{\phi, \theta}[\mu], g(\cdot, \theta) \rangle d\gamma(\theta).$$

Proposition (Backprojection, Adjoint)

$$\mathcal{R}_{\phi, \gamma}^*[h](\mathbf{x}) := \int_{\Theta} h(S_{\phi, \theta}(\mathbf{x}), \theta) d\gamma(\theta), \quad \mathbf{x} \in \mathbb{X}.$$

For essentially bounded measurable function h say $h \in L_{\lambda_{\mathbb{R} \times \gamma}}^{\infty}(\mathbb{R} \times \Theta, \mathbb{R})$ i.e.

$$\exists C > 0 \text{ s.t. } |h(t, \theta)| \leq C \text{ a.e. } (t, \theta) \in \mathbb{R} \times \Theta,$$

then

$$\langle \mathcal{R}_{\phi, \gamma}[\mu], h \rangle = \langle \mu, \mathcal{R}_{\phi, \gamma}^*[h] \rangle \quad \forall h \in L_{\lambda_{\mathbb{R} \times \gamma}}^{\infty}(\mathbb{R} \times \Theta).$$

Proposition (Affine Transformation $\mathbb{X} = \mathbb{R}^d, \Theta = \mathbb{S}_{d-1}$)

For any $\theta \in \mathbb{S}_{d-1}, \mathbf{A} \in \text{GL}(d), \mathbf{y} \in \mathbb{R}^d$, using $\phi(\mathbf{x}, \theta) := \langle \mathbf{x}, \theta \rangle$ and $\mu_{\mathbf{A}, \mathbf{y}} := (\mathbf{A} \cdot + \mathbf{y})_{\#} \mu$ holds

$$\mathcal{R}_{\theta}[\mu_{\mathbf{A}, \mathbf{y}}] = (\|\mathbf{A}^{\top} \theta\| \cdot + \langle \mathbf{y}, \theta \rangle)_{\#} \mathcal{R}_{\mathbf{A}^{\top} \theta / \|\mathbf{A}^{\top} \theta\|}[\mu]$$

Proposition (Affine Transformation $\mathbb{X} = \mathbb{R}^d, \Theta = \mathbb{R}^d$)

For any $\theta \in \mathbb{R}^d, \mathbf{Q} \in \text{O}(d), \mathbf{y} \in \mathbb{R}^d$, using $\phi(\mathbf{x}, \theta) := \|\mathbf{x} - \theta\|$, and $\mu_{s\mathbf{Q}, \mathbf{y}} := (s\mathbf{Q} \cdot + \mathbf{y})_{\#} \mu$ holds

$$\mathcal{R}_{\phi, \theta}[\mu_{s\mathbf{Q}, \mathbf{y}}] = (s \cdot)_{\#} \mathcal{R}_{\phi, \mathbf{Q}^{\top}(\theta - \mathbf{y})}[\mu].$$

Proposition (Affine Transformation $\mathbb{X} = \Theta = \text{SO}(3)$)

For any $\theta, \mathbf{R} \in \text{SO}(3)$, using $\phi(\mathbf{x}, \theta) := \arccos(\frac{\text{tr}(\theta^{\top} \mathbf{x}) - 1}{2})$, and $\mu_{\mathbf{R}} := (\mathbf{R} \cdot)_{\#} \mu$ holds

$$\mathcal{R}_{\phi, \theta}[\mu_{\mathbf{R}}] = \mathcal{R}_{\phi, \mathbf{R}^{\top} \theta}[\mu].$$

- **Aim:** Generalize the R-CDT^[11] on functions, for probability measures $\mathcal{P}(\mathbb{R}) \subset \mathcal{M}(\mathbb{R})$, similar to ^[12].
- The *cumulative distribution function* $F_\mu : \mathbb{R} \rightarrow [0, 1]$ is given by $F_\mu(t) := \mu((-\infty, t])$,
- its generalized inverse (*quantile function*) is given by

$$F_\mu^{[-1]}(t) := \inf\{s \in \mathbb{R} \mid F_\mu(s) > t\}, \quad t \in \mathbb{R}.$$

- Based on $\rho \in \mathcal{P}(\mathbb{R})$ (not give mass to atoms), define the *cumulative distribution transform* (CDT)

$$\hat{\mu} : \mathbb{R} \rightarrow \mathbb{R}, \quad \hat{\mu} := F_\mu^{[-1]} \circ F_\rho.$$

→ For convex cost $c : \mathbb{R} \rightarrow [0, \infty)$, the CDT solves the Monge–Kantorovich transportation problem on (ρ, μ) ^[13],

i.e.
$$\hat{\mu} = \arg \min_{T_{\#}\rho = \mu} \int_{\mathbb{R}} c(s - T(s)) \, d\rho(t), \quad \text{where } T : \mathbb{R} \rightarrow \mathbb{R} \text{ is measurable.}$$

→ $\mu \in \mathcal{P}_2(\mathbb{R}) = \{\mu \in \mathcal{P}(\mathbb{R}) \mid \int_{\mathbb{R}} t^2 \, d\mu(t) < \infty\}$, then

$$\hat{\mu} \in L_\rho^2(\mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid \text{measurable, } \int_{\mathbb{R}} |f(t)|^2 \, d\rho(t) < \infty\}.$$

→ For $\mu, \nu \in \mathcal{P}_2(\mathbb{R})$ the Wasserstein-2 distance^[4] fulfils

$$W_2^2(\mu, \nu) := \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^2} c(x - y)^2 \, d\pi(x, y) = \|\hat{\mu} - \hat{\nu}\|_\rho^2 := \int_{\mathbb{R}} |\hat{\mu}(t) - \hat{\nu}(t)|^2 \, d\rho(t).$$

[12] Diaz Martin, R., Medri, I. V., and Rohde, G. K. (2024) – Data representation with optimal transport, *arXiv:2406.15503*

[13] Villani, C. (2003) – Topics in Optimal Transportation,

- We define the *generalized R-CDT* by $\widehat{\mathcal{R}}_{\phi, \theta}[\mu] : \mathbb{R} \rightarrow \mathbb{R}$, $t \mapsto \widehat{\mathcal{R}}_{\phi, \theta}[\mu](t)$.
- For $\mu \in \mathcal{P}_2(\mathbb{X})$, it is $\mathcal{R}_{\phi, \theta}[\mu] \in \mathcal{P}_2(\mathbb{R})$ and consequently $\widehat{\mathcal{R}}_{\phi, \theta}[\mu] \in \mathbf{L}^2_\rho(\mathbb{R})$ and $\widehat{\mathcal{R}}_\phi[\mu] \in \mathbf{L}^2_{\rho \times \gamma}(\mathbb{R} \times \Theta)$,
- with the sliced Wasserstein distance of $\mu, \nu \in \mathcal{P}_2(\mathbb{X})$ given by

$$\|\widehat{\mathcal{R}}_{\phi, \theta}[\mu] - \widehat{\mathcal{R}}_{\phi, \theta}[\nu]\|_{\rho \times \gamma}^2 := \int_{\Theta} \int_{\mathbb{R}} |\widehat{\mathcal{R}}_{\phi, \gamma}[\mu](t, \theta) - \widehat{\mathcal{R}}_{\phi, \gamma}[\nu](t, \theta)|^2 d\rho(t) d\gamma(\theta) = \text{SW}_2^2(\mu, \nu) = \int_{\Theta} W_2^2(\mathcal{R}_{\phi, \theta}[\mu], \mathcal{R}_{\phi, \theta}[\nu]) d\gamma(\theta).$$

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- 1) **Normalization:** The *generalized NR-CDT* $\mathcal{N}_{\phi}[\mu](t, \theta) := \mathcal{N}_{\phi, \theta}[\mu](t)$, for $t \in \mathbb{R}, \theta \in \Theta$, is defined by

$$\mathcal{N}_{\phi, \theta}[\mu](t) := \frac{\widehat{\mathcal{R}}_{\phi, \theta}[\mu](t) - \text{mean}(\widehat{\mathcal{R}}_{\phi, \theta}[\mu])}{\text{std}(\widehat{\mathcal{R}}_{\phi, \theta}[\mu])}, \quad t \in \mathbb{R}.$$

-
- For **well-definiteness:** restrict to *uniformly bounded* ϕ : $|\phi(\mathbf{x}, \theta) - \phi(\mathbf{y}, \theta)| \leq M_L \quad \forall d_{\mathbb{X}}(\mathbf{x}, \mathbf{y}) \leq L, \forall \theta \in \Theta$,
 - and consider *measures* μ in $\mathcal{P}_{\phi, c}^+(\mathbb{X}) := \{\mu \in \mathcal{P}(\mathbb{X}) \mid \text{supp}(\mu) \subset \subset \mathbb{X} \wedge \exists c > 0 : \text{std}(\widehat{\mathcal{R}}_{\phi, \theta}[\mu]) \geq c \quad \forall \theta \in \Theta\}$.
- Support of $\mathcal{R}_{\phi, \theta}[\mu]$ is bounded, and by continuity means and standard derivations of $\widehat{\mathcal{R}}_{\phi, \theta}[\mu]$ exist, i.e. bounded away from zero → by construction, $\mathcal{N}_{\phi}[\mu]$ is bounded on $\text{supp}(\rho) \times \Theta$.

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- 2) **Sorting:** dimension reduction via bounded $h : \mathbf{L}^\infty(\Theta) \rightarrow \mathbb{R}$, i.e. boundedness-preserving (bounded to bounded sets):

$$h(g \circ \psi) = h(g) \quad \text{for all} \quad g \in \mathbf{L}^\infty(\Theta) \quad \text{and any bijection} \quad \psi : \Theta \rightarrow \Theta$$

-
- The *generalized h NR-CDT* $\mathcal{N}_{h, \phi}[\mu] : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$\mathcal{N}_{h, \phi}[\mu](t) := h(\mathcal{N}_{\phi}[\mu](t, \cdot)), \quad t \in \mathbb{R}, \quad \mu \in \mathcal{P}_{\phi, c}^+(\mathbb{X})$$

- It holds $\mathcal{N}_{h,\phi}[\mu] \in L_\rho^\infty(\mathbb{R})$ for $\mu \in \mathcal{P}_{\phi,c}^+(\mathbb{X})$
→ **well-definiteness** of the measure transform, i.e. feature extractor.
- For specific setting $(\mathbb{X}, \Theta, \phi)$ and $T \in \bullet_{\mathbb{X},\Theta}$ there exists $\hat{T} : \mathbb{R} \times \Theta \rightarrow \mathbb{R} \times \Theta$ s.t. $\widehat{\mathcal{R}_\phi[T_\# \mu]} = \hat{T} \circ \widehat{\mathcal{R}_\phi[\mu]}$
→ Yielding invariances in h NR-CDT-space under the specific affine transformations for choices of $(\mathbb{X}, \Theta, \phi)$.

Theorem (Separability)

For specific settings $(\mathbb{X}, \Theta, \phi)$, and $\mu \in \mathcal{P}_{\phi,c}^+(\mathbb{X})$ holds

- $\mathcal{N}_{h,\phi}[T_\# \mu] = \mathcal{N}_{h,\phi}[\mu]$ for the specific $T \in \bullet_{\mathbb{X},\Theta}$
 - each subset $\hat{\mathbb{F}}_\mu$ of $\mathbb{F}_\mu := \{T_\# \mu \mid T \in \bullet_{\mathbb{X},\Theta}\}$ reduces in h NR-CDT-space to single points sets,
 - separability of $\mathcal{N}_{h,\phi}[\hat{\mathbb{F}}_\mu]$ if $\mathcal{N}_{h,\phi}[\mu_i] \neq \mathcal{N}_{h,\phi}[\mu_j]$, $i \neq j$.
- Further transformations are:
 - TV-normalized: $h(g) = \sup_{P \in \mathbb{P}} \sum_{i=1}^{n_P} |g(\theta_{i+1}) - g(\theta_i)|$, $\Theta = \mathbb{S}_1$, with
 $\mathbb{P} = \{P = (\theta_1, \dots, \theta_{n_P+1}) \mid P \text{ is partition of } \mathbb{S}_1\}$,
 - average-normalized: $h(g) = \int_\Theta g(\theta) d\gamma(\theta)$ violating separability under arbitrary affine transformations, i.e. only single point sets if $T \in \bullet_{\mathbb{X},\Theta}$ s.t. $\kappa(T^\top T) = 1$.

- Let $\eta: \mathbb{R} \rightarrow \mathbb{R}$ be boundedness-preserving and $H: \mathcal{B}(\mathbb{R}) \rightarrow \mathbb{R}$ bounded,
- By setting $h(g) = H(\{\eta(g(\theta)) \mid \theta \in \Theta\})$ for $g \in L^\infty(\Theta)$, the required construction for h is obtained, including
 - $\eta = |\cdot|$ and $H = \sup$, i.e.,

$$\sup_{\theta \in \Theta} |\mathcal{N}_\theta[\mu](t)|,$$
 - $\eta = |\cdot|$ and $H = \inf$, i.e.,

$$\inf_{\theta \in \Theta} |\mathcal{N}_\theta[\mu](t)|,$$
 - $\eta = |\cdot|$ and $H = \sup - \inf$, i.e.,

$$\sup_{\theta \in \Theta} |\mathcal{N}_\theta[\mu](t)| - \inf_{\theta \in \Theta} |\mathcal{N}_\theta[\mu](t)|,$$
 - $\eta = \text{Id}$ and $H = \sup - \inf$, i.e.,

$$\sup_{\theta \in \Theta} \mathcal{N}_\theta[\mu](t) - \inf_{\theta \in \Theta} \mathcal{N}_\theta[\mu](t).$$

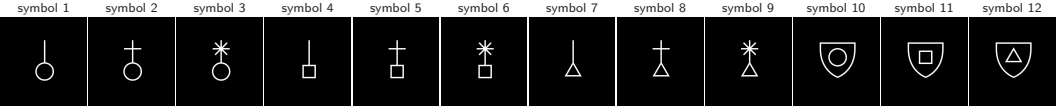


Figure: Templates for academic datasets with domain $[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}]^2$ represented by 256×256 pixels.

Table: NT classification accuracies for the academic dataset with 10 samples per class generated by random pixel shifts in $[-20, 20]$ and isotropic scaling in $[0.5, 1.25]$.

		angles		1		2		4		8		16		32		64		128		256	
				$\ \cdot\ _\infty$	$\ \cdot\ _2$	$\ \cdot\ _\infty$	$\ \cdot\ _2$	$\ \cdot\ _\infty$	$\ \cdot\ _2$	$\ \cdot\ _\infty$	$\ \cdot\ _2$	$\ \cdot\ _\infty$	$\ \cdot\ _2$	$\ \cdot\ _\infty$	$\ \cdot\ _2$	$\ \cdot\ _\infty$	$\ \cdot\ _2$	$\ \cdot\ _\infty$	$\ \cdot\ _2$	$\ \cdot\ _\infty$	$\ \cdot\ _2$
shift	R-CDT			1.0000	0.9583	1.0000	0.9583	0.9583	0.5250	0.8666	0.4750	0.9083	0.4666	0.9250	0.4583	0.9333	0.4583	0.9333	0.4500	0.9333	0.4500
	_m NR-CDT			1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9916	1.0000	0.9666	1.0000	0.9666	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	_a NR-CDT			1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	Eucl.			0.0833	0.1333																
shift & scaling	R-CDT			0.4833	0.4250	0.4833	0.4250	0.5250	0.4166	0.4583	0.3500	0.4416	0.3500	0.4333	0.3500	0.4416	0.3500	0.4416	0.3500	0.4416	0.3500
	_m NR-CDT			1.0000	0.9916	1.0000	0.9916	1.0000	1.0000	0.9833	1.0000	0.9666	1.0000	0.9666	1.0000	0.9750	1.0000	0.9750	1.0000	0.9750	1.0000
	_a NR-CDT			1.0000	0.9916	1.0000	0.9916	1.0000	1.0000	0.9916	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	Eucl.			0.0833	0.0916																

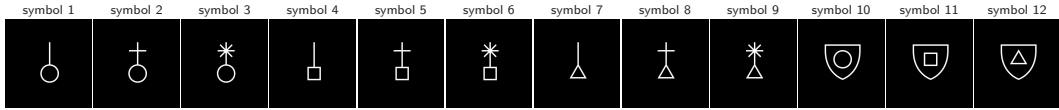


Figure: Templates for academic datasets with domain $[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}]^2$ represented by 256×256 pixels.

Table: NT classification accuracies for academic datasets with 10 samples per class and different parameter ranges for random affine transformations. Rotation angles are in $[0^\circ, 360^\circ]$, pixel shifts in $[-20, 20]$.

angles	scaling in $[0.5, 1.25]$, shearing in $[-45^\circ, 45^\circ]$						scaling in $[0.75, 1.25]$, shearing in $[-30^\circ, 30^\circ]$						scaling in $[0.75, 1.0]$, shearing in $[-15^\circ, 15^\circ]$						no scaling and shearing					
	R-CDT		mNR-CDT		NR-CDT		R-CDT		mNR-CDT		NR-CDT		R-CDT		mNR-CDT		NR-CDT		R-CDT		mNR-CDT		NR-CDT	
	ℓ_1	ℓ_2	ℓ_1	ℓ_2	ℓ_1	ℓ_2	ℓ_1	ℓ_2	ℓ_1	ℓ_2	ℓ_1	ℓ_2	ℓ_1	ℓ_2	ℓ_1	ℓ_2	ℓ_1	ℓ_2	ℓ_1	ℓ_2	ℓ_1	ℓ_2	ℓ_1	ℓ_2
1	0.0833	0.1500	0.1500	0.2666	0.1500	0.2666	0.1416	0.2250	0.1750	0.2916	0.1750	0.2916	0.1833	0.2666	0.1916	0.3083	0.1916	0.3083	0.1916	0.2666	0.1500	0.2750	0.1500	0.2750
2	0.0833	0.1500	0.1583	0.2666	0.1916	0.2666	0.1416	0.2250	0.1750	0.3000	0.2416	0.3000	0.1833	0.2666	0.1916	0.3166	0.2833	0.3166	0.1916	0.2666	0.1500	0.3000	0.2750	0.3000
4	0.0916	0.1416	0.3416	0.5000	0.2750	0.3583	0.1250	0.1416	0.3000	0.5416	0.3333	0.4333	0.1750	0.1750	0.3166	0.5666	0.3583	0.4750	0.2000	0.2333	0.3916	0.5333	0.3583	0.4583
8	0.1333	0.1583	0.5333	0.6833	0.4833	0.5666	0.1500	0.1500	0.5500	0.6416	0.5833	0.6583	0.1583	0.1916	0.5500	0.6416	0.5333	0.6000	0.2166	0.2416	0.6416	0.7416	0.5583	0.6416
16	0.1416	0.1750	0.8083	0.9250	0.8750	0.8750	0.1250	0.1583	0.8750	0.9416	0.9750	0.9833	0.1416	0.2000	0.8583	0.9000	0.9500	0.9666	0.2000	0.2833	0.8583	0.9166	0.9833	0.9666
32	0.1250	0.1583	0.9583	1.0000	0.9750	0.9583	0.1166	0.1500	0.9666	1.0000	1.0000	1.0000	0.1750	0.1833	0.9666	1.0000	1.0000	1.0000	0.2166	0.2833	0.9583	1.0000	1.0000	1.0000
64	0.1083	0.1583	0.9833	1.0000	0.9666	0.9666	0.1333	0.1500	1.0000	1.0000	1.0000	1.0000	0.1666	0.1916	1.0000	1.0000	1.0000	1.0000	0.2083	0.2833	1.0000	1.0000	1.0000	1.0000
128	0.1250	0.1583	0.9916	1.0000	0.9750	0.9666	0.1416	0.1500	1.0000	1.0000	1.0000	1.0000	0.1666	0.1916	1.0000	1.0000	1.0000	1.0000	0.2166	0.2833	1.0000	1.0000	1.0000	1.0000
256	0.1250	0.1583	0.9916	1.0000	0.9666	0.9666	0.1416	0.1416	1.0000	1.0000	1.0000	1.0000	0.1666	0.1916	1.0000	1.0000	1.0000	1.0000	0.2166	0.2833	1.0000	1.0000	1.0000	1.0000
Eucl.	0.0833	0.0916					0.0833	0.0833					0.0833	0.0666					0.0833	0.0666				
CNN-10		0.894						0.8722						0.8750						0.8720				
CNN-100		0.9153						0.9542						0.9847						0.9736				

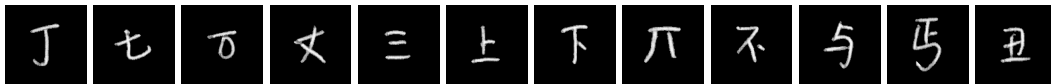
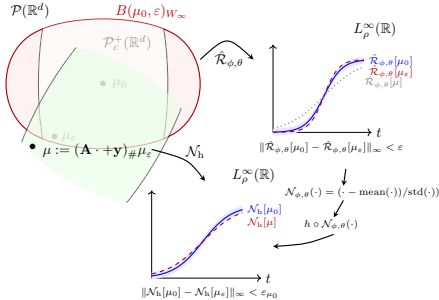


Table: NT and NN classification accuracies for the Chinese character dataset, where each class consists of 50 samples, generated by random affine transformations with scaling $[0.5, 1.0]$, shearing $[-25^\circ, 25^\circ]$, rotation angles $[0^\circ, 360^\circ]$ and pixel shifts $[-20, 20]$.

# classes	method	NT							NN	
		# angles							# training samples	
		2	4	8	16	32	64	128	5	10
100	Euclidean	0.0096							0.0202 ± 0.0021	0.0242 ± 0.0022
	R-CDT	0.0160	0.0180	0.0172	0.0164	0.0168	0.0170	0.0170	0.0188 ± 0.0020	0.0217 ± 0.0021
	_m NR-CDT	0.0924	0.1038	0.3124	0.8422	0.9836	1.0000	1.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	_a NR-CDT	0.0810	0.0172	0.1860	0.4950	0.6486	0.6814	0.6852	0.8345 ± 0.0097	0.9139 ± 0.0062
1000	_m NR-CDT	0.0848	0.0620	0.2178	0.7368	0.9791	0.9975	0.9981	0.9987 ± 0.0001	0.9987 ± 0.0001
	_a NR-CDT	0.0629	0.0362	0.0751	0.3475	0.5554	0.6030	0.6104	0.7651 ± 0.0026	0.8631 ± 0.0018



- For $\mu \in \mathcal{P}_1(\mathbb{R}^d)$ define the zero-mean quantile $\tilde{\mathcal{N}}_\theta[\mu](t) := \hat{\mathcal{R}}_\theta[\mu](t) - \text{mean}(\hat{\mathcal{R}}_\theta[\mu]) \quad \forall \theta \in \Theta$
- $\mathcal{N}_m[\mu] := \sup_{\theta \in \mathbb{S}_{d-1}} \mathcal{N}_\theta(t)$ for $\mu \in \mathcal{P}_c^*(\mathbb{R}^d)$
 $= \mathcal{P}_{c, \phi}^+(\mathbb{R}^d) = \{\mu \in \mathcal{P}(\mathbb{R}^d) \mid \text{supp}(\mu) \subset \subset \mathbb{R}^d$
 $\wedge \dim(\text{supp}(\mu)) > d-1\}.$

Proposition

Let $\mu_0, \mu_\epsilon \in \mathcal{P}(\mathbb{R}^2)$ with $W_\infty(\mu_0, \mu_\epsilon) \leq \epsilon$. Then

- i) $\|\hat{\mathcal{R}}_\theta[\mu_0] - \hat{\mathcal{R}}_\theta[\mu_\epsilon]\|_\infty \leq \epsilon$, ii) $\|\tilde{\mathcal{N}}_\theta[\mu_0] - \tilde{\mathcal{N}}_\theta[\mu_\epsilon]\|_\infty \leq 2\epsilon$.

Proposition

Let $\mu_0 \in \mathcal{P}_c^*(\mathbb{R}^d)$ and $\mu_\epsilon \in \mathcal{P}(\mathbb{R}^d)$ with $W_\infty(\mu_0, \mu_\epsilon) \leq \epsilon$, and let $c_0 := \min_{\theta \in \mathbb{S}_{d-1}} \text{std}(\hat{\mathcal{R}}_\theta[\mu_0]) > 0$. If $\epsilon < c_0/2$, then

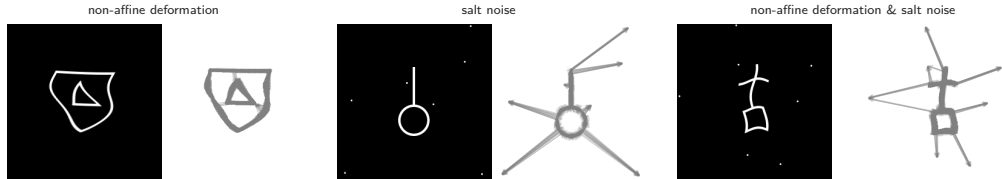
$$\|\mathcal{N}_m[(\mathbf{A} \cdot + \mathbf{y})_\# \mu_0] - \mathcal{N}_m[(\mathbf{B} \cdot + \mathbf{z})_\# \mu_\epsilon]\|_\infty \leq \frac{4(\text{diam}(\mu_0) + 2\epsilon)}{c_0(c_0 - 2\epsilon)} \epsilon.$$

Theorem (Separability with Robustness)

Let $\mu_0, \nu_0 \in \mathcal{P}_c^*(\mathbb{R}^d)$ with $\mathcal{N}_m[\mu_0] \neq \mathcal{N}_m[\nu_0]$, for $\bullet \in \{\mu, \nu\}$ let $c_\bullet := \min_{\theta \in \mathbb{S}_{d-1}} \text{std}(\hat{\mathcal{R}}_\theta[\bullet_0])$. Let $\epsilon < \frac{\min\{c_\mu, c_\nu\}}{2}$ satisfy

$$4\epsilon \left(\frac{\text{diam}(\mu_0) + 2\epsilon}{c_\mu(c_\mu - 2\epsilon)} + \frac{\text{diam}(\nu_0) + 2\epsilon}{c_\nu(c_\nu - 2\epsilon)} \right) < \|\mathcal{N}_m[\mu_0] - \mathcal{N}_m[\nu_0]\|_\infty.$$

Let $\mathbf{F}_\bullet := \{(\mathbf{A} \cdot + \mathbf{y})_\# \mid \bullet \in \mathcal{P}(\mathbb{R}^d), W_\infty(\bullet, \bullet_0) \leq \epsilon\}$. Then, $\tilde{\mathbf{F}}_\mu \subset \mathbf{F}_\mu$ and $\tilde{\mathbf{F}}_\nu \subset \mathbf{F}_\nu$ are linearly separable.



$$W_2(\rho_{\text{org}}, \rho_{\text{dis}}) = 1.9860 \cdot 10^{-3}$$

$$W_2(\rho_{\text{org}}, \rho_{\text{dis}}) = 3.1034 \cdot 10^{-3}$$

$$W_2(\rho_{\text{org}}, \rho_{\text{dis}}) = 5.77112 \cdot 10^{-3}$$

Figure: Instances of corrupted data regarding non-affine deformations and impulsive/salt noise considered in the robustness analysis. The accompanying vector fields illustrate the optimal Wasserstein-2 transport between the corrupted ρ_{dis} and true ρ_{org} .

Table: NT classification accuracies for the academic dataset with 10 samples per class and different parameter ranges for random non-affine distortions. Additionally, random affine transformations are applied with scaling in $[0.75, 1.0]$, shearing in $[-5^\circ, 5^\circ]$, rotation angles in $[0^\circ, 360^\circ]$ and pixel shifts in $[-20, 20]$.

angles	no non-affine distortion			freq. in $[0.5, 2.0]$, amp. in $[2.5, 7.5]$			freq. in $[0.5, 2.0]$, amp. in $[8.0, 13.0]$			freq. in $[0.5, 4.0]$, amp. in $[0.5, 2.0]$			freq. in $[0.5, 4.0]$, amp. in $[0.5, 7.5]$			freq. in $[0.5, 4.0]$, amp. in $[2.5, 7.5]$		
	R-CDT	μ NR-CDT	μ NR-CDT	R-CDT	μ NR-CDT	μ NR-CDT	R-CDT	μ NR-CDT	μ NR-CDT	R-CDT	μ NR-CDT	μ NR-CDT	R-CDT	μ NR-CDT	μ NR-CDT	R-CDT	μ NR-CDT	μ NR-CDT
1	0.3000	0.3250	0.3250	0.2833	0.2416	0.2416	0.2250	0.1916	0.1916	0.3000	0.2833	0.2833	0.2583	0.2166	0.2166	0.2500	0.1833	0.1833
2	0.3000	0.3250	0.3000	0.2833	0.2500	0.2583	0.2250	0.1916	0.1833	0.3000	0.3000	0.2916	0.2583	0.2166	0.2166	0.2500	0.1833	0.1833
4	0.2083	0.5833	0.5000	0.1833	0.5416	0.4250	0.1666	0.4833	0.3583	0.1916	0.5500	0.4583	0.1916	0.5000	0.3416	0.1750	0.4916	0.3166
8	0.1833	0.7166	0.6666	0.1583	0.6916	0.6166	0.1416	0.5500	0.5500	0.1750	0.7500	0.6583	0.1750	0.6166	0.5666	0.1750	0.6000	0.5250
16	0.1916	0.9083	0.9666	0.1583	0.8916	0.9416	0.1583	0.7250	0.8500	0.1750	0.8916	0.9500	0.1750	0.7833	0.8333	0.1750	0.7583	0.8000
32	0.1916	1.0000	1.0000	0.1583	0.9666	1.0000	0.1666	0.8416	0.9333	0.1833	1.0000	1.0000	0.2000	0.9083	0.9583	0.1916	0.8916	0.9500
64	0.1916	1.0000	1.0000	0.1750	0.9666	1.0000	0.1750	0.8166	0.9500	0.2000	1.0000	1.0000	0.2000	0.9250	0.9666	0.1916	0.8916	0.9416
128	0.1916	1.0000	1.0000	0.1750	0.9666	1.0000	0.1833	0.8083	0.9416	0.2000	1.0000	1.0000	0.2083	0.9333	0.9750	0.1916	0.8833	0.9500
256	0.1916	1.0000	1.0000	0.1750	0.9666	1.0000	0.1833	0.8833	1.0000	0.2000	1.0000	1.0000	0.2083	0.9333	0.9750	0.2000	0.8833	0.9666
Eucl.	0.0916			0.0583			0.0666			0.0583			0.0583			0.0583		
CNN-10	0.8903			0.8792			0.8750			0.8806			0.8861			0.8819		
CNN-100	0.9764			0.9778			0.9542			0.9750			0.9633			0.9594		

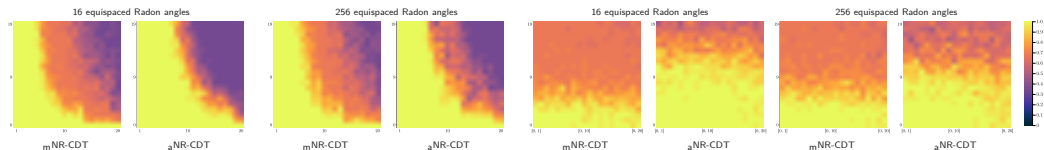


Figure: Phase transition of NT classification accuracies for 3-class academic datasets with 10 samples per class. *Left*: different salt noise (vertical: component numbers, horizontal: noise strengths); *Right*: non-affine deformations with frequencies in $[0, 2]$ and variable amplitude ranges (horizontal), as well as salt noise of strength 9 and variable location numbers (vertical).

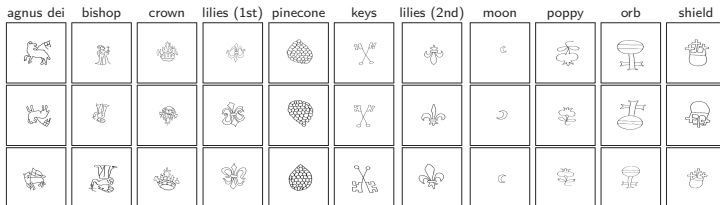
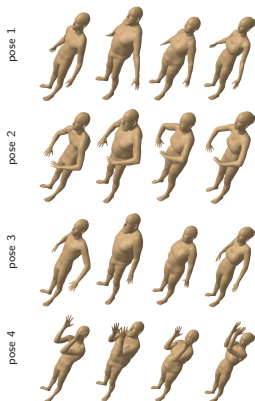


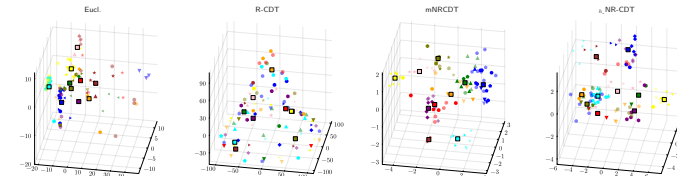
Table: NT classification accuracies for the historic watermark dataset.

# angles	Eucl.	MB	R-CDT	mNR-CDT	h _g NR-CDT	t _v NR-CDT
2			0.140	0.516	0.203	0.200
4			0.219	0.781	0.703	0.719
8			0.188	0.828	0.782	0.766
16			0.203	0.859	0.781	0.766
32			0.203	0.859	0.800	0.822
64			0.203	0.859	0.813	0.891
128			0.203	0.859	0.813	0.891
256			0.203	0.859	0.813	0.891
512			0.203	0.859	0.813	0.922
1024			0.203	0.859	0.796	0.938

human 1 human 3 human 5 human 7



class: $\bigcirc$ 1 $\star$ 2 $\triangle$ 3 $\diamond$ 4 ∇ 5 $\square$ 6 $\triangleleft$ 7 $\triangleright$ 8 $\bigcirc$ 9 $\bigcirc$ 10
 centre: $\blacksquare$ 1 $\blacksquare$ 2 $\blacksquare$ 3 $\blacksquare$ 4 $\blacksquare$ 5 $\blacksquare$ 6 $\blacksquare$ 7 $\blacksquare$ 8 $\blacksquare$ 9 $\blacksquare$ 10
 train/test: $\blacksquare$ 1 $\blacksquare$ 2 $\blacksquare$ 3 $\blacksquare$ 4 $\blacksquare$ 5 $\blacksquare$ 6 $\blacksquare$ 7 $\blacksquare$ 8 $\blacksquare$ 9 $\blacksquare$ 10



train	Eucl.	R-CDT	mNR-CDT	h_c NR-CDT
1	0.658 ± 0.054	0.475 ± 0.060	0.706 ± 0.044	0.672 ± 0.030
3	0.727 ± 0.035	0.575 ± 0.041	0.801 ± 0.050	0.784 ± 0.042
5	0.757 ± 0.037	0.591 ± 0.040	0.838 ± 0.044	0.828 ± 0.038

Table: 1-NN classification accuracies for the FAUST dataset with 10 samples per class and using 256 directions.

		Eucl.	R-CDT	mNR-CDT	h_c NR-CDT
R_{train}	($\uparrow$)	0.8261	0.8424	0.9208	0.9208
V_{train}	($\downarrow$)	1.7299	2.7545	0.8316	0.9300
R_{test}	($\uparrow$)	0.7910	0.8424	0.9346	0.9444
V_{test}	($\downarrow$)	1.6587	2.7964	0.6975	0.7074

Table: Quality measures for 10-means clustering of the FAUST dataset with 10 samples per class, where 5 samples are used for training and the rest for testing.

Included Publications

- [1] Beckmann, M., Beinert, R., and Bresch, J. Max-normalized Radon CDT for limited data classification. In: *Lect. Notes Comput. Sci.* Lect. Notes Comput. Sci. 15667 (2025), pp. 241–254. DOI: 10.1007/978-3-031-92366-1_19.
- [2] Beckmann, M., Beinert, R., and Bresch, J. Normalized Radon CDT for invariance and robustness in optimal transport based image classification. In: *SIAM J. Imaging Sci.* 19 (2 2026). DOI: 10.1137/25M1767741.
- [3] Beckmann, M., Beinert, R., and Bresch, J. Generalization of the normalized Radon CDT for limited data recognition. In: *J. Math. Imaging Vis.* 68.45 (2026). DOI: 10.1007/s10851-026-01327-4.

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- [5] Hauser, D. et al. “On image processing and pattern recognition for thermograms of watermarks in manuscripts – A first proof-of-concept.” In: *International Conference on Document Analysis and Recognition (ICDAR)*. Vol. 14806. Lect. Notes Comput. Sci. Athens, Greece: Springer, 2024, pp. 91–107. DOI: 10.1007/978-3-031-70543-4_6.
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Objects as Prob. Measures

- Classification with *many classes* and *few samples*
- Assumption: (approx.) *affine* / structured class families
- Need: a) *invariance* to affine transformations
b) *stability* to deformations
- Application: historic watermarks, automated pattern recognition



Normalization and Sorting

- For $\mu \in \mathcal{P}_{\phi, c}^+(\mathbb{X}) \subset \mathcal{P}_c(\mathbb{X}) \subset \mathcal{P}_2(\mathbb{X})$ R-CDT is bounded with

$$\text{std}(\widehat{\mathcal{R}_{\phi, \theta}[\mu]}) \geq c > 0 \quad \text{f.a.} \quad \theta \in \Theta.$$
- Normalize per slice:**

$$\mathcal{N}_{\phi, \theta}[\mu](t) = \frac{\widehat{\mathcal{R}_{\phi, \theta}[\mu]}(t) - \text{mean}(\widehat{\mathcal{R}_{\phi, \theta}[\mu]})}{\text{std}(\widehat{\mathcal{R}_{\phi, \theta}[\mu]})}$$
- Sorting** the directions $\theta \in \Theta$ via:

$$h : L^\infty(\Theta) \rightarrow \mathbb{R}$$
- Define** ${}_h\text{NR-CDT}$: $h(\mathcal{N}_{\phi}[\mu](t, \cdot))$

Radon on Measures & Core Idea

- Restricted Radon transform via measurable $\phi : \mathbb{X} \times \Theta \rightarrow \mathbb{R}$

$$\mathcal{R}_{\phi, \theta} : \mathcal{M}(\mathbb{X}) \rightarrow \mathcal{M}(\mathbb{R})$$

$$\mu \mapsto (S_{\phi, \theta})_{\#} \mu, \quad S_{\phi, \theta} := \phi(\cdot, \theta)$$

- Slice per direction

$$\mu \in \mathcal{P}(\mathbb{X}) \Rightarrow \mathcal{R}_{\phi, \theta}[\mu] \in \mathcal{P}(\mathbb{R})$$

- 1D embedding via CDT, wrt. ref ρ

$$\hat{\mu} = \arg \min_{T_{\#} \rho = \mu} \int_{\mathbb{R}} c(s - T(s)) d\rho(s), \quad \text{atomless } \rho, \mu \in \mathcal{P}(\mathbb{R})$$

- R-CDT: collect sliced CDTs

$$\widehat{\mathcal{R}_{\phi, \theta}[\mu]} \in L_{\rho}^2(\mathbb{R}) \quad \text{for } \mu \in \mathcal{P}_2(\mathbb{X})$$

Theoretical Results

- Translation:** for suitable $(\mathbb{X}, \Theta, \phi)$

$$\widehat{\mathcal{R}_{\phi}[T_{\#} \mu]} = \hat{T} \circ \widehat{\mathcal{R}_{\phi}[\mu]}$$

for specific T correspond to $\hat{T}$

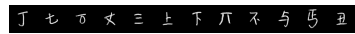
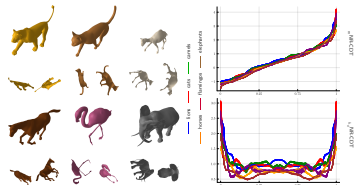
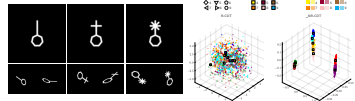
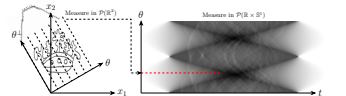
- i) **Invariance:** $\mathcal{N}_{h, \phi}[T_{\#} \mu] = \mathcal{N}_{h, \phi}[\mu]$

- ii) **Singleton sets** in ${}_h\text{NR-CDT}$ -space

$$\mathbb{F}_{\mu} = \{T_{\#} \mu\} \xrightarrow{{}_h\text{NR-CDT}} \mathcal{N}_{h, \phi}[\mathbb{F}_{\mu}] = \mathcal{N}_{h, \phi}[\mu]$$

- iii) **Robustness** of ${}_m\text{NR-CDT}$ and ${}_a\text{NR-CDT}$ in Wasserstein-spaces

$$h \in \{\sup_{\theta \in \Theta} \cdot, \int_{\Theta} \cdot\}.$$



# lines	method	NT # angles	16	32	64	128	1-NN # training samples
		8					5
E	Euclidean	0.0096					0.0202 ± 0.0021
	R-CDT	0.0102	0.0188	0.0190	0.0190	0.0192	0.0178 ± 0.0022
	${}_m\text{NR-CDT}$	0.4052	0.9032	0.9022	0.9992	1.0000	0.0210 ± 0.0024
E	${}_m\text{NR-CDT}$	0.3433	0.8256	0.9842	0.9977	1.0000	0.9987 ± 0.0001
							0.9997 ± 0.0001
# angles		8	16	32	64	128	
		ℓ_1	ℓ_1	ℓ_1	ℓ_1	ℓ_1	
		ℓ_2	ℓ_2	ℓ_2	ℓ_2	ℓ_2	
R-CDT		0.4285	0.1785	0.3928	0.1785	0.4642	0.1785
		0.8571	0.9642	0.9285	1.0000	0.9285	0.4285
		0.9642	0.8571	1.0000	0.9642	1.0000	0.9642
${}_m\text{NR-CDT}$		0.9642	0.8571	1.0000	0.9642	1.0000	0.9642
		0.8571	0.9642	0.9285	1.0000	0.9285	0.9642
		0.9642	0.8571	1.0000	0.9642	1.0000	0.9642
Eucl.		0.2857	0.1785				